BSIT 375: Knowledge Discovery and Data Science

Week 3 Lecture Notes: Preparing to Model the Data

Based on Larose & Larose: *Discovering Knowledge in Data (2nd Ed)*

Instructor: Tek Raj Pant

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Introduction

Before building predictive or descriptive models, it is essential to understand the foundational principles that guide model selection, evaluation, and performance. This week focuses on key concepts that prepare us for effective data modeling: the distinction between supervised and unsupervised learning, the interplay of statistical and data mining methodologies, techniques for model validation, the dangers of overfitting, the bias–variance trade-off, handling imbalanced datasets, and the importance of establishing baseline performance.

1 Supervised versus Unsupervised Methods

Data mining tasks are broadly categorized into supervised and unsupervised learning.

Supervised Learning

In supervised learning, the dataset includes a **target variable** (also called label or response) that we aim to predict. The algorithm learns a function that maps input features to the target based on example input–output pairs.

- **Goal:** Prediction (classification or regression).
- **Examples:** Predicting whether a customer will churn (yes/no), forecasting house prices, diagnosing disease from medical images.
- **Algorithms:** Linear regression, logistic regression, decision trees, random forests, support vector machines, neural networks.

Unsupervised Learning

Here, no target variable exists. The algorithm searches for hidden patterns, groupings, or structures within the data.

- **Goal:** Description, pattern discovery, dimensionality reduction.
- **Examples:** Customer segmentation, anomaly detection, market basket analysis.
- **Algorithms:** k-means clustering, hierarchical clustering, principal component analysis (PCA), association rules (Apriori).

Key Distinction

Supervised methods use labelled data to *predict* an outcome; unsupervised methods explore unlabelled data to *discover* hidden structures.

2 Statistical Methodology and Data Mining Methodology

Though both statistics and data mining deal with data analysis, their philosophies and typical workflows differ.

Statistical Methodology

- **Hypothesis-driven:** Begins with a hypothesis, collects data to test it.
- **Explanatory focus:** Aims to understand relationships (e.g., "Is variable X significant?").
- **Population inference:** Often focuses on inference from sample to population.
- **Model parsimony:** Prefers simpler models that are interpretable.

Data Mining Methodology

- **Discovery-driven:** Exploratory approach, seeks patterns without a priori hypotheses.
- **Predictive focus:** Accuracy of predictions is paramount, even if the model is complex.
- **Big data:** Often deals with massive datasets where computational efficiency matters.
- **Validation emphasis:** Relies heavily on out-of-sample validation (e.g., cross-validation).

Modern data science blends both: using statistical thinking to avoid spurious patterns while leveraging data mining techniques for high predictive power.

3 Cross-Validation

Cross-validation is a resampling technique used to assess how a model generalizes to an independent dataset. It helps detect overfitting and guides model selection.

Holdout Method

The simplest approach: split the data into a **training set** (e.g., 70%) and a **test set** (30%). Train the model on the training set, evaluate on the test set. *Problem:* The evaluation may be highly sensitive to the split.

k-Fold Cross-Validation

To reduce variability, k-fold cross-validation:

1. Partition the data into k equal-sized folds.
2. For each fold $i = 1, \ldots, k$:
   - Train the model on all folds except fold i.
   - Evaluate on fold i (the validation set).
3. Compute the average performance across all k folds.

Common choices: $k = 5$ or $k = 10$. When $k = n$ (where n is the number of observations), we obtain **leave-one-out cross-validation** (LOOCV).

Stratified k-Fold

For classification, it is important to preserve the class distribution in each fold. **Stratified k-fold** ensures each fold has roughly the same proportion of classes as the whole dataset.

Cross-Validation Formula

For a loss function L (e.g., squared error, misclassification rate), the cross-validation estimate is:

$$CV = \frac{1}{k} \sum_{i=1}^{k} \frac{1}{|D_i|} \sum_{(x,y) \in D_i} L(y, \hat{f}^{-i}(x))$$

where $\hat{f}^{-i}$ is the model trained without fold i, and D_i is the i-th fold.

4 Overfitting

Definition

Overfitting occurs when a model learns not only the underlying pattern but also the noise and random fluctuations in the training data. Such a model performs exceptionally well on training data but poorly on unseen test data.

Causes

- Model too complex (e.g., too many parameters, deep trees).
- Insufficient training data relative to model complexity.
- Lack of regularization.

Detection

- Compare training error vs. validation error: if training error is much lower than validation error, overfitting is present.
- Learning curves: plot training and validation error as a function of training set size or model complexity.

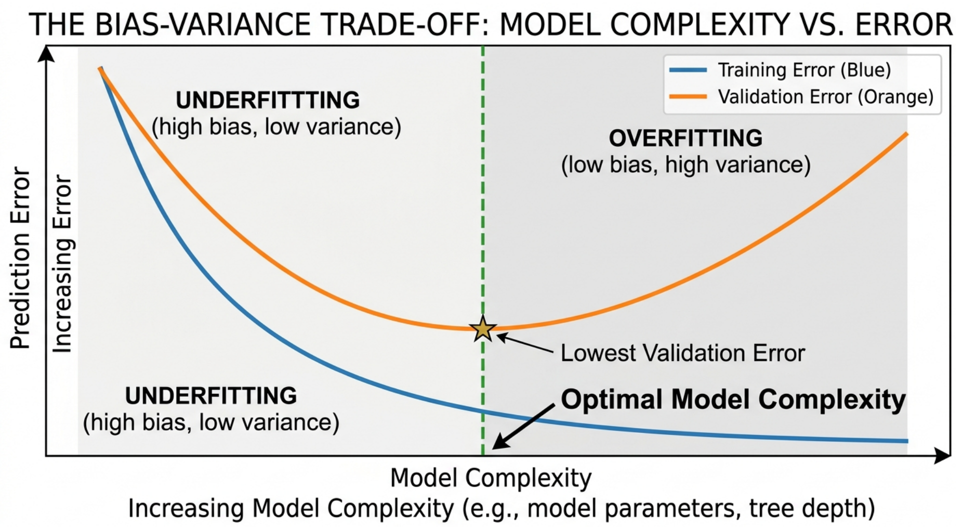

Figure 1: Typical overfitting behavior: validation error increases after a certain complexity threshold.

5 Bias–Variance Trade-off

The bias–variance trade-off is a fundamental concept explaining the error of a model.

Mathematical Decomposition

For a regression setting with squared error loss, the expected prediction error at a point x can be decomposed as:

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Var} + \sigma^2$$

where:

- Bias^2: error due to incorrect assumptions (underfitting).
- Var: error due to sensitivity to small fluctuations in the training set (overfitting).
- σ^2: irreducible error (noise inherent to the data).

Interpretation

- **Simple models** (e.g., linear regression) have high bias but low variance.
- **Complex models** (e.g., deep trees) have low bias but high variance.
- The optimal model balances bias and variance to minimize total error.

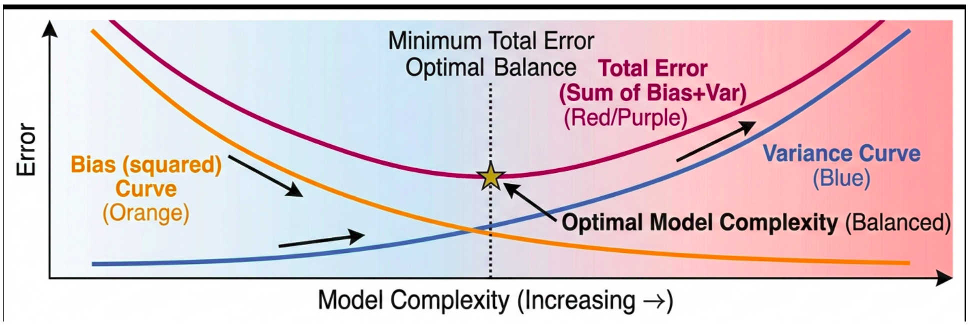

Figure 2: The bias–variance trade-off: increasing model complexity reduces bias but increases variance.

6 Balancing the Training Data Set

Many real-world classification problems involve **imbalanced classes**—one class (the majority) vastly outnumbers another (the minority).

Why Balance Matters

Standard classifiers tend to be biased toward the majority class, leading to high overall accuracy but poor performance on the minority class. For example, in fraud detection, if 99% of transactions are legitimate, a model that always predicts "legitimate" achieves 99% accuracy but catches no fraud.

Techniques to Address Imbalance

1. **Random Undersampling**: Remove random instances from the majority class until balance is achieved.
2. **Random Oversampling**: Duplicate random instances from the minority class.
3. **SMOTE** (Synthetic Minority Over-sampling Technique): Create synthetic minority samples by interpolating between existing minority instances.
4. **Cost-sensitive Learning**: Assign higher misclassification costs to the minority class.

Evaluation Metrics for Imbalanced Data

Accuracy can be misleading. Instead, use:

- Precision, Recall, F1-score.
- Area Under the ROC Curve (AUC).
- Confusion matrix.

7 The Confusion Matrix

The Confusion Matrix is a fundamental tool for visualizing the performance of a supervised learning algorithm.

Table 1: Binary Confusion Matrix

	Predicted: Positive	**Predicted: Negative**
Actual: Positive	True Positive (TP)	False Negative (FN)
Actual: Negative	False Positive (FP)	True Negative (TN)

8 Core Classification Metrics

Precision (Exactness)

Precision measures the accuracy of positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

Recall (Sensitivity / Completeness)

Recall measures the ability of the model to find all the positive units in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

F1-Score (Harmonic Mean)

The F1-Score provides a balance between Precision and Recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

9 ROC and AUC

The **Receiver Operating Characteristic (ROC)** curve is a plot of the True Positive Rate (TPR) against the False Positive Rate (FPR) at various thresholds.

- **True Positive Rate (Recall):** $TPR = \frac{TP}{TP+FN}$
- **False Positive Rate:** $FPR = \frac{FP}{FP+TN}$

The **Area Under the Curve (AUC)** is the integral of the ROC curve:

$$AUC = \int_0^1 TPR(FPR)\, dFPR \tag{4}$$

An AUC of 1.0 represents a perfect classifier, while an AUC of 0.5 represents random guessing.

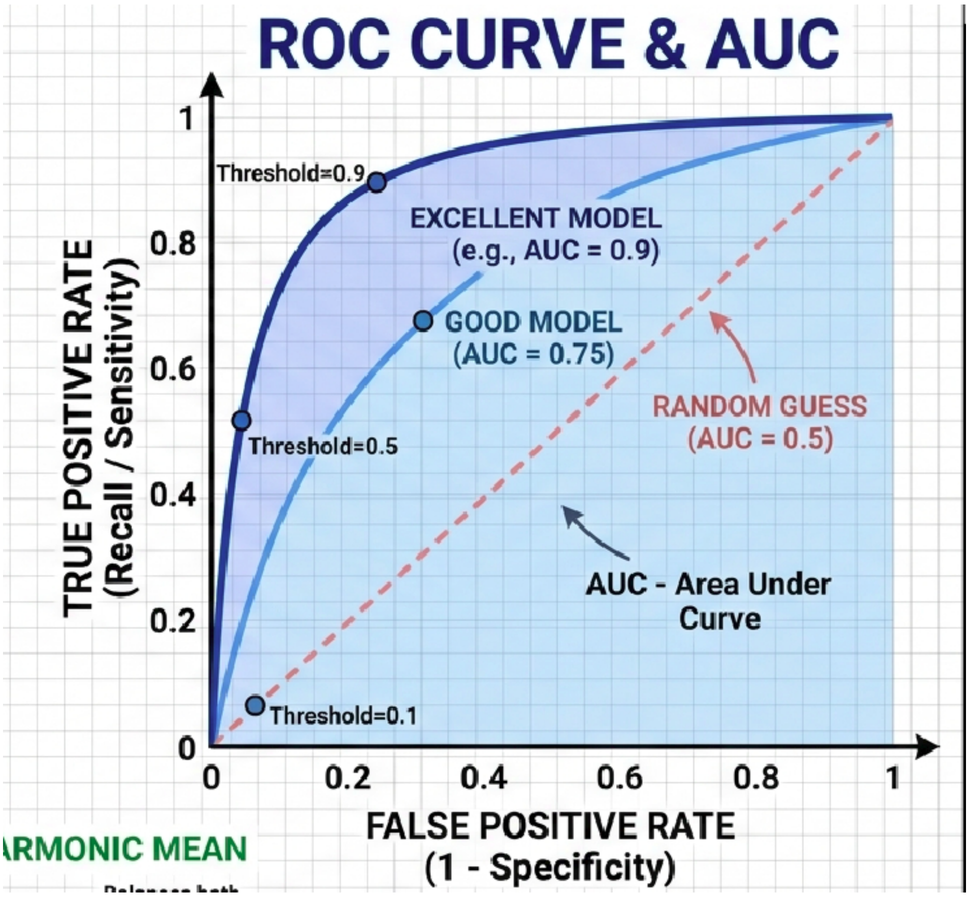

Figure 3: ROC Curve

10 Numerical Example

Consider a test set of $N = 100$ samples. The distribution of actual and predicted classes is summarized in the following confusion matrix:

Table 2: Classification Results for $N = 100$

	Predicted: Positive	**Predicted: Negative**	**Total**
Actual: Positive	30 (TP)	10 (FN)	40
Actual: Negative	5 (FP)	55 (TN)	60
Total	35	65	100

11 Performance Metrics Calculation

Using the values from the confusion matrix, we calculate the primary evaluation metrics as follows:

Precision

Precision measures the proportion of positive identifications that were actually correct:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 5} = \frac{30}{35} \approx 0.857\ (85.7\%) \tag{5}$$

Recall (Sensitivity)

Recall measures the proportion of actual positives that were identified correctly:

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{30}{30 + 10} = \frac{30}{40} = 0.750\ (75.0\%) \tag{6}$$

F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a single balanced metric:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{0.857 \cdot 0.750}{0.857 + 0.750} \approx 0.800\ (80.0\%) \tag{7}$$

12 Receiver Operating Characteristic (ROC) and AUC

The ROC curve plots the **True Positive Rate (TPR)** against the **False Positive Rate (FPR)**.

- **True Positive Rate (TPR):** $\frac{TP}{TP+FN} = 0.750$
- **False Positive Rate (FPR):** $\frac{FP}{FP+TN} = \frac{5}{5+55} \approx 0.083$

The **Area Under the Curve (AUC)** quantifies the overall ability of the model to discriminate between classes. Mathematically, it is the area under the ROC curve $f(x)$:

$$AUC = \int_0^1 TPR(FPR)\, dFPR \tag{8}$$

13 The ROC Function and Numerical Integration

For our specific numerical example, the function $TPR(FPR)$ is defined as a piecewise linear curve passing through the points $(0, 0)$, $(0.083, 0.750)$, and $(1, 1)$. The integration of this function yields the AUC:

$$AUC = \int_0^{0.083} 9.036x\, dx + \int_{0.083}^{1} (0.273x + 0.727)\, dx \tag{9}$$

Calculating the definite integrals:

- **Area 1:** $[4.518x^2]_0^{0.083} = 0.0311$
- **Area 2:** $[0.1365x^2 + 0.727x]_{0.083}^{1} = 0.8024$

$$\textbf{Total AUC} = \mathbf{0.8335} \tag{10}$$

This represents the probability that a randomly selected positive instance will be ranked higher than a randomly selected negative instance by the classifier.

14 Establishing Baseline Performance

Before building complex models, it is critical to define a **baseline**—a simple model or heuristic against which to compare.

Common Baselines

- **Majority Class Classifier**: Always predict the most frequent class.
- **Random Classifier**: Predict randomly according to class distribution.
- **Simple Rule**: e.g., "if feature is greater than threshold then class A else class B".
- For regression: predict the mean of the target variable.

Purpose

- **Context:** Know what "good" performance means.
- **Sanity check:** If a sophisticated model does not beat the baseline, it is not useful.
- **Resource allocation:** Avoid over-engineering when a simple model suffices.

Example

In a churn prediction problem with 10% churn rate, a baseline that always predicts "no churn" achieves 90% accuracy. Any model must exceed this baseline in a meaningful way (e.g., by improving recall for the churn class).